

Binomial and Negative Binomial Distributions

Definitions, Comparison, and Worked Examples

Department of Computer Science

University of Botswana

September 21, 2026



Learning Objectives

By the end of this lecture, you should be able to:

- identify when a binomial model is appropriate;
- identify when a negative binomial model is appropriate;
- define the random variable and its possible values;
- calculate probabilities using the correct formula;
- calculate the mean and variance;
- explain the difference between fixed trials and fixed successes.

Big Picture

Both distributions are based on repeated **Bernoulli trials**.

Bernoulli trial

A trial with only two outcomes:

$$\text{success} \quad \text{or} \quad \text{failure}.$$

Examples

- Toss a coin: heads = success, tails = failure.
- Test a component: defective = success, not defective = failure.
- Login attempt: successful login = success, failed login = failure.

Starting Point: Bernoulli Trials

Both binomial and negative binomial distributions use the same basic conditions.

Conditions

- Each trial has two outcomes: success or failure.
- Trials are independent.
- The probability of success is constant: $\mathbb{P}(\text{success}) = p$.
- The probability of failure is $q = 1 - p$.

Example

Repeatedly toss a coin and call heads a success. If the coin is fair, then

$$p = \mathbb{P}(\text{heads}) = 0.5, \qquad q = 0.5.$$

The Central Difference

	Binomial	**Negative binomial**
What is fixed?	Number of trials, n	Number of successes, r
What is random?	Number of successes	Number of trials needed
Typical wording	"in 10 trials"	"until the 3rd success"
Possible values	$0, 1, \ldots, n$	$r, r+1, r+2, \ldots$

Memory aid

Binomial: stop after a fixed number of trials.
Negative binomial: stop when a fixed number of successes has occurred.

Which Model Should I Use?

Ask this question first

What does the problem fix?

Problem wording	**Model**
"Out of n trials, how many successes?"	Binomial
"In n attempts, exactly x successes"	Binomial
"How many trials until r successes?"	Negative binomial
"The rth success occurs on trial x"	Negative binomial

Simple rule

Fixed trials $\Rightarrow$ Binomial. Fixed successes $\Rightarrow$ Negative binomial.

Binomial Distribution: Definition

Let X be the number of successes in n independent trials, where each trial has success probability p.

$$X \sim \text{Bin}(n, p).$$

Probability mass function

$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

- n is fixed.
- x is the number of successes.
- $n - x$ is the number of failures.

Understanding the Binomial Formula

$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

Meaning of each part

- $\binom{n}{x}$ chooses the positions of the x successes.
- p^x gives the probability of the x successes.
- $(1-p)^{n-x}$ gives the probability of the $n-x$ failures.

Example wording

"A test has 10 questions. Find the probability of exactly 7 correct answers."

This is binomial because the number of trials is fixed at 10.

Binomial Distribution: Mean and Variance

If

$$X \sim \text{Bin}(n, p),$$

then

$$\mathbb{E}[X] = np, \qquad \text{Var}(X) = np(1-p).$$

The standard deviation is

$$\sigma = \sqrt{np(1-p)}.$$

Quick interpretation

If $n = 20$ and $p = 0.30$, then

$$\mathbb{E}[X] = 20(0.30) = 6.$$

Over many groups of 20 trials, the average number of successes will be about 6.

Worked Example 1: Exactly x Successes

Question

A student answers 10 independent true-or-false questions. The probability of a correct answer is 0.60. Find the probability of exactly 7 correct answers.

Let

$$X = \text{number of correct answers}, \qquad X \sim \text{Bin}(10, 0.60).$$

$$\mathbb{P}(X = 7) = \binom{10}{7}(0.60)^7(0.40)^3.$$

$$\mathbb{P}(X = 7) = 120(0.60)^7(0.40)^3 \approx 0.2150.$$

Worked Example 2: Cumulative Probability

Question

A component fails a test with probability 0.10. Ten components are tested independently. Find the probability that at most two fail.

Let

$$X \sim \text{Bin}(10, 0.10).$$

"At most two" means $X \leq 2$.

$$\mathbb{P}(X \leq 2) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1) + \mathbb{P}(X = 2).$$

$$\mathbb{P}(X \leq 2) = \sum_{x=0}^{2} \binom{10}{x} (0.10)^x (0.90)^{10-x} \approx 0.9298.$$

Negative Binomial Distribution: Definition

Let X be the trial number on which the rth success occurs.

$$X \sim \text{NegBin}(r, p).$$

Probability mass function

$$\mathbb{P}(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \qquad x = r, r+1, r+2, \ldots$$

- r is fixed.
- X is random because we do not know how many trials will be needed.

Why $\binom{x-1}{r-1}$?

Suppose the rth success occurs on trial x.

Key idea

The last trial, trial x, must be a success.

Therefore, in the first $x-1$ trials, we must have exactly $r-1$ successes.

$$\binom{x-1}{r-1}$$

chooses where those first $r-1$ successes occurred.

Memory aid

Do not use $\binom{x}{r}$. The final trial is already fixed as a success.

Negative Binomial: Two Common Conventions

Textbooks use two different random variables. Always define yours.

	Trials until rth success	**Failures before rth success**
Random variable	X	Y
Values	$x = r, r+1, \ldots$	$y = 0, 1, 2, \ldots$
Formula	$\binom{x-1}{r-1} p^r q^{x-r}$	$\binom{y+r-1}{r-1} p^r q^y$
Relationship	$Y = X - r$, so $X = Y + r$	

Convention used in these slides

X counts the total number of trials needed for r successes.

Negative Binomial: Mean and Variance

If X counts the total number of trials needed for r successes, then

$$\mathbb{E}[X] = \frac{r}{p}, \qquad \mathrm{Var}(X) = \frac{r(1-p)}{p^2}.$$

If $Y = X - r$ counts failures before the rth success, then

$$\mathbb{E}[Y] = \frac{r(1-p)}{p}, \qquad \mathrm{Var}(Y) = \frac{r(1-p)}{p^2}.$$

Notice

Subtracting the constant r changes the mean but does not change the variance.

Worked Example 3: The rth Success

Question

A basketball player makes each free throw with probability 0.40. Find the probability that the player's third successful shot occurs on the seventh attempt.

Here

$$r = 3, \qquad x = 7, \qquad p = 0.40.$$

$$\mathbb{P}(X = 7) = \binom{7-1}{3-1}(0.40)^3(0.60)^{7-3}.$$

$$\mathbb{P}(X = 7) = \binom{6}{2}(0.40)^3(0.60)^4 = 0.124416.$$

Worked Example 4: More Than k Trials

Question

A network packet is transmitted successfully with probability 0.80. What is the probability that more than 5 attempts are needed to obtain 3 successful transmissions?

More than 5 attempts are needed exactly when the first 5 attempts contain at most 2 successes. Let

$$Z \sim \text{Bin}(5, 0.80).$$

$$\mathbb{P}(X > 5) = \mathbb{P}(Z \leq 2) = \sum_{z=0}^{2} \binom{5}{z} (0.80)^z (0.20)^{5-z}.$$

$$\mathbb{P}(X > 5) = 0.00032 + 0.00640 + 0.05120 = 0.05792.$$

Connection to the Geometric Distribution

The geometric distribution is a special case of the negative binomial distribution.

When $r = 1$

If X counts the trial number on which the first success occurs, then

$$X \sim \text{Geometric}(p).$$

$$\mathbb{P}(X = x) = (1 - p)^{x-1} p, \qquad x = 1, 2, 3, \ldots$$

Meaning

Geometric asks: "How many trials until the first success?"

Comparison of the Formulas

	Binomial	**Negative binomial**
Model	$X \sim \text{Bin}(n, p)$	$X \sim \text{NegBin}(r, p)$
Fixed quantity	n trials	r successes
PMF	$\binom{n}{x} p^x q^{n-x}$	$\binom{x-1}{r-1} p^r q^{x-r}$
Mean	np	r/p
Variance	npq	rq/p^2
Values	$0 \leq x \leq n$	$x \geq r$

Which Distribution Should Be Used?

1. A factory checks 20 items and records the number that are defective.
   Binomial: the number of trials is fixed at 20.
2. A factory checks items until it finds 4 defective items.
   Negative binomial: the number of successes is fixed at 4.
3. A player takes 12 penalty kicks and records the number scored.
   Binomial: the number of kicks is fixed at 12.
4. A player continues taking penalties until scoring 5 goals.
   Negative binomial: the target is 5 successes.

Practice Questions

1. A fair die is rolled 12 times. Find the probability of obtaining exactly three sixes.
2. A computer login succeeds with probability 0.70 on each independent attempt. Find the probability that the fourth successful login occurs on the sixth attempt.
3. For Question 2, find the expected number of attempts needed to obtain four successful logins.
4. State the appropriate distribution:
    1. the number of defective chips in a sample of 30;
    2. the number of chips tested before the fifth defective chip is found.

Practice Answers

1. $X \sim \text{Bin}(12, 1/6)$

$$\mathbb{P}(X = 3) = \binom{12}{3} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^9 \approx 0.1974.$$

2. $X \sim \text{NegBin}(4, 0.70)$

$$\mathbb{P}(X = 6) = \binom{5}{3} (0.70)^4 (0.30)^2 \approx 0.2161.$$

3.

$$\mathbb{E}[X] = \frac{r}{p} = \frac{4}{0.70} \approx 5.71 \text{ attempts.}$$

4. (a) Binomial. (b) Negative binomial.

Common Mistakes

- Using the binomial distribution when the trial count is not fixed.
- Forgetting that the last trial in a negative binomial problem must be a success.
- Using $\binom{x}{r}$ instead of $\binom{x-1}{r-1}$ when X counts total trials.
- Mixing up "at most" ($\leq$) and "at least" ($\geq$).
- Failing to state whether the negative binomial variable counts trials or failures.
- Using the model when trials are dependent or p changes between trials.

Summary

Binomial

Fix n trials and count successes:

$$\mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}.$$

Negative binomial

Fix r successes and count trials until the rth success:

$$\mathbb{P}(X = x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}.$$

Key message

The wording of the problem tells you which quantity is fixed.